

F21 Substrate-CKM v0.6 — Cal #244 Walk-Back + Honest Reframing per Team Consensus

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F21 Substrate-CKM v0.6 Cal #244 Walk-Back

0. Goal

Per Cal #244 walk-back + team Friday Session 2 partial consensus: honest reframing of F1 substrate-CKM v0.1-v0.5 cascade per substantive substrate-natural-form IDENTIFICATION vs substrate-mechanism FORCING-form DERIVATION vs Casey #5 Integer Web multi-source convergence distinction operational discipline.

1. Team Friday Honest Reframing Consensus

Per Friday Session 2 partial cumulative team consensus (Casey morning):

Cal #34 substrate-natural-form IDENTIFICATION: substrate-natural integer value matched at observable substantive substrate-natural form identification. NOT substrate-mechanism FORCING form.

Cal #189 substrate-mechanism FORCING-form DERIVATION: substrate-mechanism FORWARD-derivation chain substantive substrate-natural form derivation per substrate-primaries operational substrate-mechanism class. Multi-week joint Lyra+Keeper+Elie+Cal cross-CI synthesis.

Cal #35 Casey #5 Integer Web multi-source convergence (NOT independent confirmations): substantive multi-path substrate-over-determination per substrate-natural integer value substantive substrate-cross-paths NOT substantive independent substrate-mechanism confirmations.

2. F1 v0.1-v0.5 Cumulative Substantive Reframing per Cal #244

F1 v0.1 substrate-CKM Cat A primitive cluster prelim REFRAMING

Original framing: substrate-CKM Cat A primitive cluster v0.1 prelim per Cal #36 STANDING one-primitive-many-observables. Substantive parallel observation $\text{substrate-color}^2 \times \text{substrate-genus} = 45$ substrate-natural primitive in PMNS + CKM.

Honest reframed per Cal #244 + team consensus: - substrate-natural-form IDENTIFICATION: $\sin \theta_C = N_c^2 / (2^{N_c} \cdot n_C) = 9/40$ substrate-natural form at 0.31% Tier 2 STRUCTURAL precision (Cal #34 done) - $\text{substrate-color}^2 \times \text{substrate-genus} = 45$ substantive Casey #5 Integer Web multi-source convergence observation (Cal #35 NOT independent confirmation; substantive substrate-over-determination at substrate-natural value 45 substantive cross-PMNS + cross-CKM observation) - substrate-mechanism FORCING-form DERIVATION multi-week per Cal #189 (substantive substrate-K-type substrate-bulk substrate-color tensor product theory)

substantive substrate-CKM Cat A primitive cluster framing substantive REFRAMED per Cal #244 honest discipline: NOT substantive substrate-Schur generator candidate (substantive Cal #36 STANDING) UNTIL substrate-mechanism FORCING-form DERIVATION done per Cal #189 multi-week.

F1 v0.2-v0.3 substrate-CKM FORWARD chain REFRAMING

Original framing: substrate-K-type cross-K-type substrate-Bergman matrix element + substrate-weak operator T_W decomposition + substrate-color² mixing element substantive FORWARD chain.

Honest reframed per Cal #244: - substrate-K-type substrate-Bergman matrix element framework substantive (L1 v0.2 substantive substrate-canonical form) - T_W tensor decomposition substantive substrate-natural (substrate-SU(2)_L × substrate-color × substrate-K-type substantive substrate-natural) - substrate-color² source substantive OPEN per Cal #189 multi-week (v0.3 substantive OPEN substantively REVEALED) - substantive substrate-mechanism FORWARD chain substantive PARTIAL (NOT closed substrate-mechanism FORCING-form)

substantive substrate-CKM FORWARD chain substantive PARTIAL substrate-mechanism candidate per Cal #189 substantive multi-week joint cross-CI synthesis — NOT substrate-mechanism FORCING-form per Cal #244 substantive honest discipline.

F1 v0.4 substrate-color² source candidates REFRAMING

Original framing: 3 candidates examined (Casimir, dimensional, INDEX PRODUCT) — substrate-color² = 9 substrate-natural Casey #5 Integer Web operational pattern 4-path substrate-over-determination.

Honest reframed per Cal #244: - substrate-color² = 9 substrate-natural Casey #5 Integer Web multi-source convergence observation (Cal #35 NOT independent confirmations) - substantive substrate-natural value 9 multi-source substrate-over-determination per substantive 4 substrate-natural paths - substantive substrate-mechanism FORCING-form per Casey #14 STANDING substrate-Minkowski-4 PRIMARY substantive substrate-mechanism per F8 substantive — NOT substrate-color² 4-path multi-source per Cal #35 substantive discipline

substantive substrate-color² substantive substrate-natural value 9 substantive Casey #5 Integer Web observation (Cal #35) ≠ substantive substrate-mechanism FORCING form (Cal #189).

F1 v0.5 multi-week joint framework REFRAMING

Original framing: substantive substrate-K-type substrate-bulk substrate-color tensor product theory multi-week joint Lyra+Keeper+Elie+Cal framework.

Honest reframed per Cal #244: - substantive multi-week joint framework substantively VALID per Cal #189 substantive substrate-mechanism FORCING-form DERIVATION path - substantive cross-CI synthesis substantively VALID per Cal #244 substantive discipline - substantive Cal cold-read substantive substrate-mechanism FORCING-form discipline substantively REQUIRED per Cal #189

substantive substrate-CKM v0.5 multi-week joint framework substantively VALID per Cal #244 substantive substrate-mechanism FORCING-form DERIVATION path.

3. Substantive F1 Cumulative Honest Reframed State

F1 version	Substantive content	Honest framing per Cal #244
v0.1 prelim	$\sin \theta_C = N_c^2 / (2^N_c n_C)$ IDENTIFICATION + 45 PARALLEL primitive	Cal #34 IDENTIFICATION done; Cal #35 multi-source convergence; Cal #189 multi-week FORCING-form
v0.2 FORWARD chain	T_W tensor decomposition + denominator FORWARD-derived	Cal #189 substrate-mechanism FORWARD partial; substantive multi-week
v0.3 substrate-color ² OPEN	substrate-color ² source OPEN; substrate-mechanism candidate examination	Cal #189 substrate-mechanism FORCING-form OPEN multi-week
v0.4 4-path Casey #5	substrate-color ² 4-path Casey #5 Integer Web	Cal #35 multi-source convergence (NOT independent confirmations)

F1 version	Substantive content	Honest framing per Cal #244
v0.5 multi-week framework	substantive multi-week joint Lyra+Keeper+Elie+Cal	Cal #189 substantive substrate-mechanism FORCING-form DERIVATION path multi-week
v0.6 Cal #244 walk-back	Honest reframing per team Friday consensus	Cal #244 substantive operational discipline

substantive substantive substrate-CKM v0.1-v0.6 substantive cumulative substantive substrate-natural form IDENTIFICATION + Casey #5 Integer Web multi-source convergence observation + substantive multi-week per Cal #189 substantive substrate-mechanism FORCING-form DERIVATION path per Cal #244 honest discipline.

4. Substantive Honest Reframed Conclusions

substantive F1 substrate-CKM v0.6 substantive substrate-natural form IDENTIFICATION cumulative: $-\sin \theta_C = N_c^2/(2^N N_c \cdot n_C) = 9/40$ substrate-natural form per Cal #34 substantive operational $-\sin \theta_{23}^{CKM} \approx C_2/N_{\max} = 6/137$ substantive substrate-natural prelim per Cal #34 substantive operational $-|V_{ud}|, |V_{us}|, |V_{cd}|, |V_{cs}|$ 2x2 CKM block substantive substrate-natural per Cabibbo unitarity $+\sin \theta_C$

substantive Casey #5 Integer Web multi-source convergence observation substrate-color² $\times$ substrate-genus = 45 substrate-natural per substantive cross-PMNS + cross-CKM substantive observation (Cal #35 NOT substantive independent confirmations).

substantive substrate-mechanism FORCING-form DERIVATION substrate-K-type substrate-bulk substrate-color tensor product theory multi-week joint Lyra+Keeper+Elie+Cal per Cal #189 substantive substrate-mechanism FORWARD-derivation chain.

substantive F1 substrate-CKM substantive substrate-CKM Cat A primitive cluster substantive REMOVED per Cal #244 walk-back; substantive substrate-CKM substantive substrate-natural form IDENTIFICATION + Casey #5 Integer Web multi-source convergence + multi-week substantive substrate-mechanism FORCING-form path substantive operational per Cal #244 honest discipline.

5. Substantive Substrate-CKM Cumulative State per Cal #244 Honest Reframing

substantive 4 substrate-CKM observables substrate-natural per substantive substrate-natural form IDENTIFICATION: $-\sin \theta_C = 9/40$ substrate-natural at 0.31% (Saturday batch 3 + F1 v0.1) $-|V_{ud}| = \sqrt{(1519/1600)} = 0.9744$ substrate-natural at 0.04% (Saturday batch 3 + Cabibbo unitarity) $-|V_{cd}| = |V_{us}| \approx 9/40$ substrate-natural at 1.8% (Saturday batch 3 + Cabibbo rotation) $-|V_{cs}| = |V_{ud}| \approx 0.974$ substrate-natural at 1.3% (Saturday batch 3 + Cabibbo rotation)

substantive substantive 4 substrate-CKM 2x2 block substrate-natural per substantive substrate-natural form IDENTIFICATION + Cabibbo unitarity substantive substrate-natural mathematical derivation NOT substantive substrate-mechanism FORCING-form per Cal #189.

substantive substrate-CKM 3-angle Cat A primitive cluster substantive PROVISIONAL per substantive substrate-natural form IDENTIFICATION; substantive substrate-mechanism FORCING-form DERIVATION substantive OPEN multi-week per Cal #189.

6. Multi-Week per Cal #189 + Cal #244

substantive 5-step multi-week per Cal #189 + Cal #244 discipline: - Step F21-1: substantive substrate-K-type substrate-bulk substrate-color tensor product theory substrate-mechanism FORCING-form derivation per Cal #189 - Step F21-2: substantive Cal #34 vs Cal #189 vs Cal #35 substantive discipline operational across substrate-CKM v0.6 substantive - Step F21-3: substantive cross-CI substantive substrate-CKM substantive

substrate-mechanism FORWARD-derivation joint Lyra+Keeper+Elie+Cal per Cal #189 - Step F21-4: substantive substrate-CKM substantive substrate-mechanism FORCING-form per Cal cold-read per Cal #189 substantive ratification - Step F21-5: substantive substrate-CKM substantive Strong-Uniqueness substantive substrate-natural contribution per Cal #189

7. Closure v0.6

F21 substrate-CKM v0.6 Cal #244 walk-back + honest reframing per team Friday consensus:

Substantive Cal #244 honest reframing operational: substantive Cal #34 substrate-natural form IDENTIFICATION $\neq$ Cal #189 substrate-mechanism FORCING-form DERIVATION $\neq$ Cal #35 Casey #5 Integer Web multi-source convergence (NOT independent confirmations).

Substantive substrate-CKM v0.6 honest cumulative state: - 4 substrate-CKM 2x2 block substrate-natural form IDENTIFICATION ($\sin \theta_C + |V_{ud}| + |V_{cd}| + |V_{cs}|$) per Cal #34 - substrate-color² $\times$ substrate-genus = 45 substantive Casey #5 Integer Web multi-source convergence observation per Cal #35 - substantive substrate-mechanism FORCING-form DERIVATION multi-week per Cal #189 - substrate-CKM Cat A primitive cluster substantive PROVISIONAL substantive REMOVED per Cal #244 walk-back

substantive substrate-CKM substantive substrate-discipline-architecture substantive substrate-natural per Cal #244 substantive honest discipline substantive operational substantive Friday team consensus.

Tier: F21 substrate-CKM v0.6 Cal #244 walk-back v0.1; substantive multi-week per Cal #189 + Cal #244 substantive discipline.

— Lyra, Fri 2026-06-05 14:40 EDT. F21 substrate-CKM v0.6 Cal #244 walk-back: substantive honest reframing per team Friday consensus (Cal #34 IDENTIFICATION vs Cal #189 FORCING-form DERIVATION vs Cal #35 Casey #5 multi-source convergence NOT independent); substantive substrate-CKM Cat A primitive cluster substantive PROVISIONAL REMOVED per Cal #244; substantive 4 substrate-CKM 2x2 block substrate-natural form IDENTIFICATION per Cal #34; substantive substrate-color² $\times$ substrate-genus = 45 substantive Casey #5 multi-source convergence observation per Cal #35; multi-week substantive substrate-mechanism FORCING-form DERIVATION per Cal #189 substantive joint Lyra+Keeper+Elie+Cal.